Slide 1: AI/NLP Financial Risk Intelligence & Tactical Allocation

S&P Global & CRISIL Campus Hackathon 2026 | Case Study Round 3

- Candidate: Shivam Pal | S&P Hackathon Finalist

- Primary Deliverable: Central AI/NLP Risk Engine parsing unstructured financial news & social streams.

- Dual Downstream Implementation: Module A (Tactical Index Rebalancer) + Module B (Wholesale Portfolio Stress Tester).

- Institutional Standards: Loughran-McDonald sentiment weighting, Basel III CET1 metrics, and Bloomberg-grade dashboard.

Slide 2: Business Problem & Solution Approach

Bridging the Gap Between Unstructured Market Noise and Actionable Risk Signals

- The Challenge: Critical risk events occur in seconds across news/social feeds; manual analysis creates costly latency.

- Our Innovation: Multi-source ingestion (News + Tweets) transformed into structured machine-readable risk signals.

- Tri-Factor Risk Extraction: Sentiment Score (-1 to 1), Event Taxonomy (6 Classes), and Impact Severity (1 to 10).

- Downstream Execution: Instant index rebalancing (Module A) and automatic wholesale portfolio stress simulation (Module B).

Slide 3: End-to-End System Design & Architecture

Modular, Decoupled & High-Throughput Pipeline

- Ingestion Layer: Dual channels (Financial News + FinTwit/StockTwits) with ticker & cashtag normalization.

- Core NLP Risk Engine: Loughran-McDonald sentiment modifier, Bayesian event classification, and systemic severity scoring.

- Module A (Tactical Rebalancer): Dynamic quadratic weight optimization for S&P 100 constituent universe.

- Module B (Wholesale Stress Tester): Event-triggered ($500M book: Loans, Bonds, Swaps, CDS) macro shock simulation.

- Access Layer: High-performance FastAPI REST endpoints + Streamlit Dark Mode executive dashboard.

Slide 4: Implementation Highlights & Technical Choices

Rigorous Financial Engineering & Machine Learning

- FinBERT & Loughran-McDonald Lexicon: Heavy weighting for debt defaults, covenants, rating downgrades, and buybacks.

- Bayesian Event Classifier: Covers Geopolitical, Credit Event, Macroeconomic, M&A, and Product Launch categories.

- Impact Severity Formula: Combines systemic event weight, sentiment polarization, entity prominence, and urgency tokens.

- Risk Constraints (Module A): Minimum single-stock floor of 2.0% and maximum cap of 18.0% to prevent concentration risk.

- Wholesale Valuation (Module B): Modified Duration & Convexity (Bonds), DV01 (Swaps), CS01 (CDS), and PD/LGD Migration.

Slide 5: Key Results & Quantitative Findings

Backtested Alpha Generation & Portfolio Resilience

- Alpha Generation: Tactical Index generated +1.67% excess return over benchmark S&P 500 ETF in 30-day backtest.

- Risk-Adjusted Efficiency: Sharpe Ratio of 0.97 vs 0.72 benchmark with a controlled portfolio turnover of 16.7%.

- Stress Test Impact: Successfully simulated $11.77M MtM loss (3.38%) on $500M wholesale book under Geopolitical shock.

- Capital Telemetry: Monitored CET1 ratio decline from 13.0% to 9.71%, flagging regulatory buffer erosion under stress.

- Execution Latency: Complete signal generation and downstream rebalancing executed in under 60 milliseconds.

Slide 6: Domain Impact & Business Value

Empowering S&P Global, CRISIL & Institutional Asset Managers

- CRISIL Credit Intelligence: Early-warning detection of counterparty rating downgrades and loan default triggers.

- S&P Dow Jones Indices: Foundation for dynamic sentiment-overlay indices delivering systematic alpha.

- Wholesale Banking Stress Testing: Automated CCAR / Basel III regulatory compliance and capital adequacy monitoring.

- Operational Scalability: Fully accessible via REST API for straight-through integration into Trading & Risk desks.

Slide 7: Limitations, Assumptions & Future Roadmap

Path to Enterprise Production Deployment

- Current Assumptions: Synthetic wholesale book ($500M); simulated historical price paths with calibrated drift.

- Multilingual Intelligence: Expand to multi-lingual transformers for cross-border EMEA and APAC news coverage.

- Direct Market Execution: Integrate with FIX Protocol for straight-through order routing into execution venues.

- Agentic Explanations: Embed LLM chain-of-thought rationale alongside numerical impact severity scores.